

Worksheet: Chemical equilibrium N₂O₄ dissociation

Name(s) _____

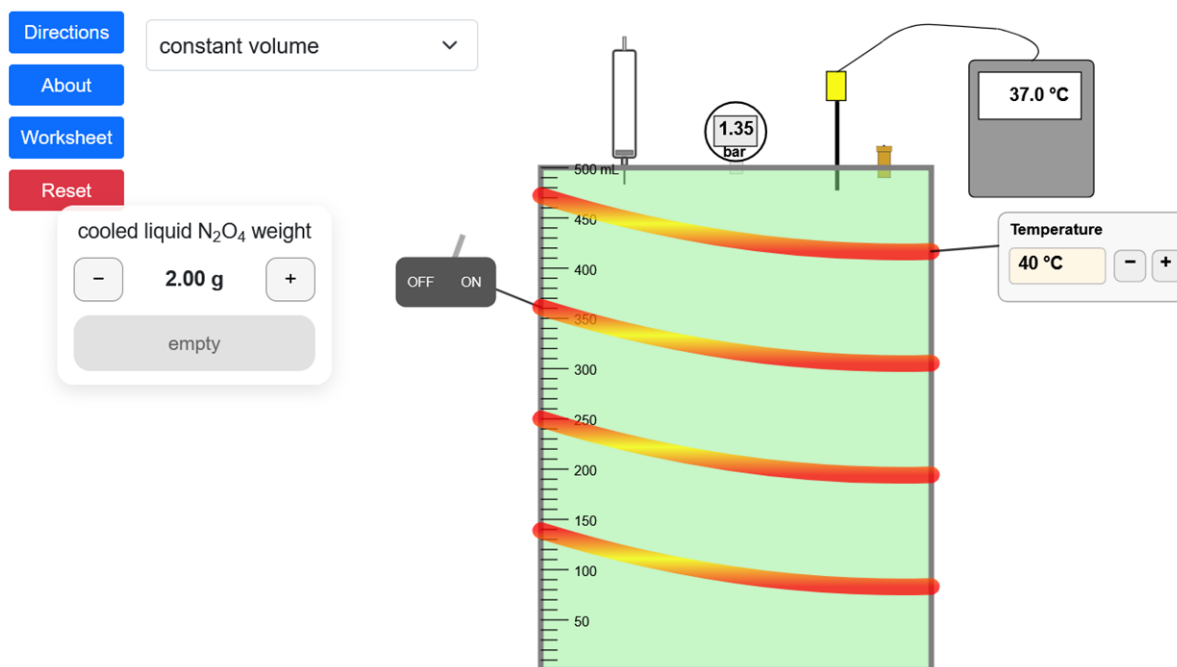
This experiment measures equilibrium pressures in a constant-volume system and equilibrium volumes in a constant-pressure system for a gas-phase reaction with a stoichiometric mole change (N₂O₄ → 2 NO₂). Equilibrium constants calculated from measurements at a range of temperatures are used to calculate the heat of reaction and the entropy change of reaction.

Student learning objectives

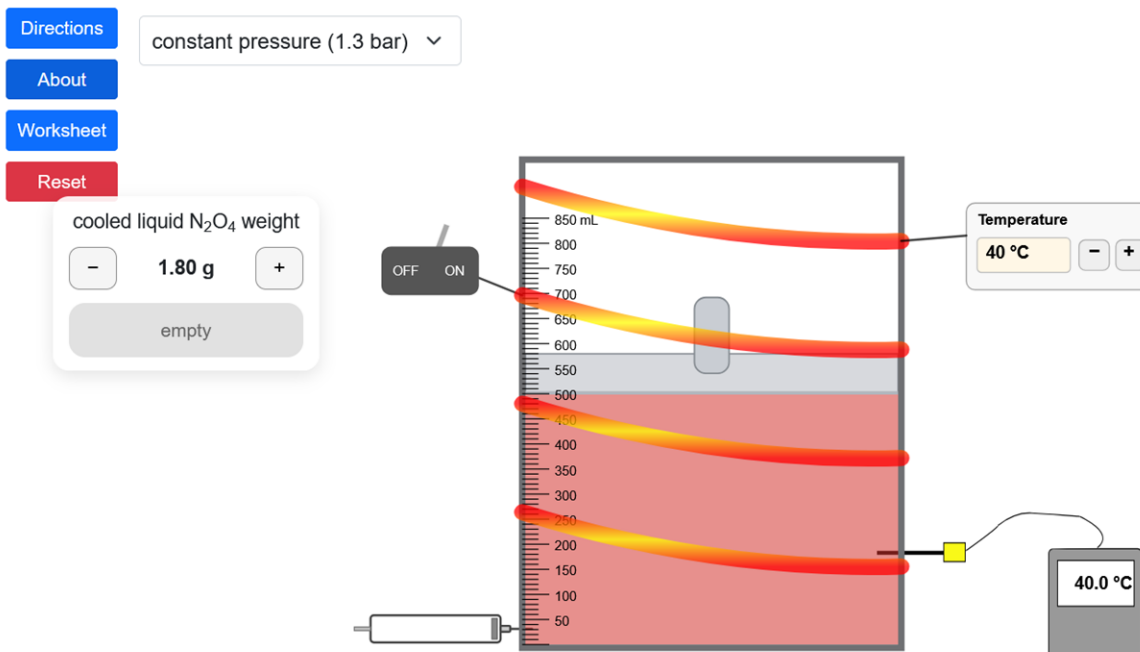
1. Apply mole balances and the ideal gas law to reacting systems at equilibrium.
2. Determine the extent of reaction from equilibrium pressures or volumes for a gas-phase reaction with a stoichiometric mole change.
3. Calculate equilibrium constants from pressures, volumes, and extents of reaction.
4. Use the van't Hoff equation to estimate ΔH° and ΔS°
5. Explain how equilibrium shifts with temperature, pressure, and volume.

Equipment

Constant volume reactor with a pressure gauge that reads absolute pressure, a heating sleeve, and a temperature sensor. A syringe injects liquid N₂O₄ into the reactor. The reactor contains a pressure-relief valve to prevent over pressure of the reactor.



Constant pressure reactor with a piston, a heating sleeve, and a temperature sensor. A syringe injects liquid N_2O_4 into the reactor.



Questions to answer before starting the experiment

1. If you add an inert gas at constant temperature to the gas-phase reaction $\text{A} \rightarrow 2\text{B}$ that is at equilibrium in a constant volume reactor, which direction does equilibrium shift? Why?
2. If you add an inert gas at constant temperature to the gas-phase reaction $\text{A} \rightarrow 2\text{B}$ that is at equilibrium in a constant pressure reactor, which direction does equilibrium shift? Why?
3. For the gas-phase reaction $\text{A} \rightarrow 2\text{B}$ at constant temperature in a piston/cylinder, when the pressure increases, which direction does the reaction proceed? Why?

Notations

$n_{N_2O_4e}$ = number of moles of N_2O_4 at equilibrium

n_{NO_2e} = number of moles of NO_2 at equilibrium

P = pressure

V = volume

R = ideal gas constant [0.08314 (bar-L)/(mol K) or 8.314 J/(mol K)]

T = temperature (absolute)

$n_{N_2O_4i}$ = number of moles of N_2O_4 injected into container

X = extent of reaction

K_{eq} = equilibrium constant (dimensionless)

$\hat{a}_{NO_2}^2$ = activity of NO_2

$\hat{a}_{N_2O_4}$ = activity of N_2O_4

$\hat{f}_i$ = fugacity of component i in mixture

f_i^o = fugacity of pure component i at 1-bar pressure

$P_{NO_2}^2$ = partial pressure of NO_2

$P_{N_2O_4}$ = partial pressure of N_2O_4

y_{NO_2} = mole fraction of NO_2

$y_{N_2O_4}$ = mole fraction of NO_2

ΔS_{rxn}^o = entropy change of reaction at standard state (1 bar and system temperature)

ΔH_{rxn}^o = heat of reaction at standard state (1 bar and system temperature)

Experiments

System 1: Equilibrium at constant volume

1. In a constant-volume reactor, the pressure increases because the number of moles increases as N_2O_4 dissociates to reach equilibrium. The pressure increases more at higher temperatures because of the relation between pressure and temperature (ideal gas law) and because equilibrium conversion increases with temperature.
2. Inject 1.6 to 2.0 g of cooled liquid N_2O_4 into an evacuated 500-mL container at 25°C. Record the amount of N_2O_4 injected _____ g and calculate the moles injected _____ mol.
3. Allow time for the N_2O_4 to completely evaporate and for the reaction to reach chemical equilibrium and record the pressure in Table 1 on page 6.

4. Calculate the total number of moles ($n_{N_2O_4e} + n_{NO_2e}$) at equilibrium from the ideal gas law. Consider using a spreadsheet for the calculations since the same calculations are done multiple times.

$$P = \frac{(n_{N_2O_4e} + n_{NO_2e})RT}{V}$$

where P = pressure

R = ideal gas constant = 0.08314 (bar-L)/(mol K)

T = temperature (K)

V = reactor volume = 0.50 L

5. Since $n_{N_2O_4e} = n_{N_2O_4i} - X$ and $n_{NO_2e} = 2X$, then

$$(n_{N_2O_4e} + n_{NO_2e}) = (n_{N_2O_4i} - X + 2X) = (n_{N_2O_4i} + X),$$

where X is the extent of reaction (units of mol). Calculate the extent of reaction from the total number of moles and the moles of N_2O_4 injected and record in Table 1.

$$X = n_{N_2O_4i} - n_{N_2O_4e}$$

where $n_{N_2O_4i}$ = mole of N_2O_4 injected

$n_{N_2O_4e}$ = moles of N_2O_4 at equilibrium

$n_{NO_2e} = 2X$ = moles of NO_2 at equilibrium

6. Calculate the equilibrium constant and record the value in Table 1.

$$K_{eq} = \frac{\hat{a}_{NO_2}^2}{\hat{a}_{N_2O_4}} \text{ where } \hat{a}_i \text{ is activity of component } i \text{ and is equal to the ratio } = \frac{\hat{f}_i}{f_i^o}$$

where $\hat{f}_i$ is the fugacity (bar) of component i in a mixture and f_i^o (bar) is the fugacity of pure component i at 1.0 bar pressure. Thus, activity is dimensionless and the equilibrium constant is dimensionless. For ideal gases, the fugacity of component i is equal to its partial pressure.

$$K_{eq} = \frac{P_{NO_2}^2}{P_{N_2O_4}} = \frac{(y_{NO_2}P)^2}{y_{N_2O_4}P} = \frac{\left(\frac{2X}{n_{N_2O_4i} + X}\right)^2 P^2}{\left(\frac{n_{N_2O_4i} - X}{n_{N_2O_4i} + X}\right) P} = \frac{4X^2 P}{(n_{N_2O_4i} + X)(n_{N_2O_4i} - X)}$$

Note that for this equation to be valid, gases must be ideal and pressure must be in bar units.

7. Turn on the heater and raise the temperature to 35°C, allow time for the reaction to equilibrate, repeat steps 3 to 6, and record values in Table 1.

8. Repeat step 7 for temperature of 45°C, 55°C, 65°C, and 75°C.

Table 1 Equilibrium constant as function of temperature

Temperature (°C)	Pressure (bar)	X (mol)	K _{eq}	Temperature (K)
25				298
35				308
45				318
55				328
65				338
75				348

System 2: Equilibrium at constant pressure

1. In a constant-pressure reactor, the volume increases because the number of moles increases as N₂O₄ dissociates to reach equilibrium. The volume increases more at higher temperatures because of the relation between volume and temperature (ideal gas law) and because equilibrium conversion increases with temperature.
2. Inject 1.6 to 2.0 g of cooled liquid N₂O₄ into an empty piston/cylinder at 25°C. The pressure exerted by the piston is 1.3 bar. Record the amount of N₂O₄ injected _____ g and calculate the moles injected _____ mol.
3. Allow time for N₂O₄ to completely evaporate and for the reaction to reach chemical equilibrium at constant pressure and measure the volume and record in Table 2.
4. Calculate the total number of moles ($n_{N_2O_4e} + n_{NO_2e}$) at equilibrium from the ideal gas law. Consider using a spreadsheet for the calculations since the same calculations are done multiple times.

5. Calculate the extent of reaction X from the total number of moles ($n_{N_2O_4i} + X$) and the moles of N_2O_4 injected and record in Table 2 on page 8.

$$P = \frac{(n_{N_2O_4e} + n_{NO_2e})RT}{V}$$

where P = pressure

R = ideal gas constant = 0.08314 (bar-L)/(mol K)

T = temperature (K)

V = reactor volume = 0.50 L

6. Calculate the equilibrium constant and enter the value in Table 2.

$$K_{eq} = \frac{4X^2P}{(n_{N_2O_4i} + X)(n_{N_2O_4i} - X)}$$

7. Turn on the heater and raise the temperature to 35°C, repeat steps 3 – 6 and record the values in Table 2.
8. Repeat step 7 for temperatures of 45°C, 55°C, 65°C, and 75°C.

Table 2 Equilibrium constant as function of temperature

Temperature (°C)	Volume (L)	X (mol)	K _{eq}	Temperature (K)
25				298
35				308
45				318
55				328
65				338
75				348

Analysis

Use the equilibrium constants in Tables 1 and 2 to calculate the heat of reaction (ΔH_{rxn}^o) and the entropy change of reaction (ΔS_{rxn}^o) from a plot of $\ln K_{eq}$ versus $1/T$ and the van't Hoff equation:

$$\ln K_{eq} = -\frac{\Delta H_{rxn}^o}{RT} + \frac{\Delta S_{rxn}^o}{R}$$

where R = ideal gas constant (J/mol K)). The van't Hoff equation assumes the heat of reaction is independent of temperature.

$$\Delta H_{rxn}^o = \underline{\hspace{2cm}}$$

$$\Delta S_{rxn}^o = \underline{\hspace{2cm}}$$

Questions to answer

1. Do equilibrium measurements at constant volume and constant pressure give similar values of K_{eq} ? Why or why not?
2. How close are your calculated values for ΔH_{rxn} and ΔS_{rxn} to values calculated from heats of formation and entropies in the NIST Webbook?
3. Does the system reach equilibrium faster at low or high temperatures? Why?
4. Why does adding an inert gas have no effect on equilibrium composition at constant volume, but shifts equilibrium at constant pressure for a gas-phase reaction with mole change?
3. What are the safety concerns with conducting this reaction? Are N_2O_4 or NO_2 toxic?
4. What assumptions in these measurements might affect the accuracy of the equilibrium constants measured?